Swapnith Varma

Portfolio Website

swapnithvarma0@gmail.com | +91-9573756869 | linkedin.com/in/swapnith-varma-ba86b4163

Education

- **B.Tech, Electronics and Communication Engineering**, JNTUH — CGPA: 9.05/10 **2016 – 2020**

Summary

Software Engineer with almost 5 years of experience specializing in C C++ and CAD–PLM integrations. Proven record of optimizing enterprise systems (70–80% performance boosts) and delivering scalable ERP–CAD and cloud-based solutions across global deployments. Recognized for quickly grasping new requirements and adapting to dynamic customer needs, with a strong curiosity that drives continuous learning.

Skills

- **Programming Languages**: C++, C#, SQL, Baan 4GL, JavaScript, Python
- **Frameworks/Systems**: .NET, Infor LN , React, Node.js, REST APIs
- **Tools & IDEs**: Git, SVN, JIRA, Amazon Q,Docker, Eclipse, Visual Studio
- **Learning/Exposure**: LLMs, NLP Preprocessing, Attention Mechanisms

Experience

- **Infor** []

Software Engineer (Jun 2024 – Present) , Associate Software Engineer (Apr 2021 – May 2024) Hyderabad, India

Enhancements in Office Integration to PLM Cloud *C#*

- Integrated Infor PLM Cloud with Microsoft applications by leveraging Microsoft APIs to enable seamless saving of Office files (Word, Excel, PowerPoint) from OneDrive to PLM Cloud.
- Developed batch register functionality to efficiently upload and manage large volumes of files through Microsoft APIs.
- Enabled multiple customers to efficiently centralize and manage their diverse Office files in the cloud.

Custom Defined Fields From CAD Integration *C#,C++, SQL, 4GL*

- Designed a logic-driven solution to import ERP-defined custom fields into CAD Integration, enabling design-specific customizations and seamless ERP–CAD synchronization, widely used by all customers.

Performance Optimization in CAD-PLM Integration *C#, C++, SQL, 4GL*

- Optimized CAD–PLM integrations (70–80% faster saves) by enhancing multi-threaded Toolkit and server-side logic, applying concurrency and fault-tolerant design principles to ensure reliable, large-scale data processing in PLM Cloud.
- Designed and implemented JSON-based request handling to manage multiple interlinked database save operations and applied concurrency concepts such as multi-threading, synchronization, and request orchestration.

Feature Enablement for ERP Concepts in CAD Integration *C#, C++*

- Adapted and extended CAD Integration Toolkit to support Create and Freeze Generation, Item Segmentation, and Custom Life Cycle functionalities by aligning with backend ERP logic.

Integration of Vertex, Oracle AutoVue Viewer with Infor PLM Cloud *C#, SQL, 4GL, Javascript,Java*

- Integrated Vertex CAD Viewer with PLM Cloud using REST APIs, designing a C proof-of-concept for file uploads and retrievals, then implementing production integration in Infor LN for 2D/3D previews, markups, and seamless synchronization.
- Integrated Oracle AutoVue Desktop Viewer with PLM Cloud for file download, markup creation, and saving back via a custom button-based interface.
- Delivered seamless CAD visualization and review for on-premise and cloud customers, now actively used in production environments.

Creo Integration API Migration *C++*

◦ Upgraded Creo integration from v8 to v9+ by adapting to new API structures, rewriting functionalities, and validating across CAD operations, ensuring continued compatibility and adoption by enterprise customers.

Multi Language Mapping in CAD-PLM Cloud Integration *C++, C#, SQL,4GL*

◦ Implemented multi-language mapping where users could define Mapping rules in multiple languages which will save data across multiple languages in a single click.

◦ Actively used by customers, saving time by eliminating the need to switch languages, thereby improving efficiency.

CAD Design Variant Handling in PLM Cloud *C++, C#, SQL, 4GL*

◦ Enhanced Creo–PLM integration by implementing variant-aware saving functionality using STL that creates separate documents for each design variant with linked neutral files, while maintaining synchronized relationships with the generic document.

◦ This was a customer-critical requirement for go-live ensuring on-time implementation by overcoming major technical challenges.

CAD Utility from Creo and Load Balancer Implementation *C++, C#, SQL, 4GL*

◦ Enhanced the CAD Utility (an executable that automates background tasks through workflows) by extending support for Creo integration. The CAD Utility automatically retrieves records, opens associated CAD files, generates neutral files, performs save operations, and closes without user intervention aligning with DevOps/SRE automation principles.

◦ Implemented a load-balancing mechanism to intelligently distribute tasks across connected systems, reducing system load and improving efficiency while enhancing availability and reliability of integrations aligning with high-availability and distributed systems principles.

Export Import of Mapping Templates across Different Tenants *Infor SQL, 4GL*

◦ Developed a feature to export CAD Mapping Templates with all associated rules and linked tables, enabling seamless one-click import into other tenants. Eliminated the need for lengthy data migration processes, significantly reducing setup time and effort for customers across multiple environments.

- Developed and enhanced features across Inventor and SolidWorks by extending Toolkit (C#,C++), database tables, and library logic (SQL, 4GL), while also resolving complex customer-reported defects through advanced debugging and root-cause analysis to ensure timely delivery.
- Worked in **Agile** methodology with monthly sprints and bi-annual releases, delivering multiple customer-focused features within tight timelines.
- Collaborated with cross-functional teams to gather requirements and communicate effective solutions, leveraging Jira for sprint planning and issue tracking to ensure alignment across development and departmental stakeholders.
- Assisted newly joined and existing team members in resolving issues and clarifying requirements, promoting smoother collaboration and knowledge sharing.

Projects

Lightweight Software Updater & Release Manager *C#, GIT*

- Built a C#-based light weight auto-update system using GitHub Releases and metadata-driven version tracking, enabling fully automated patch delivery, secure ZIP extraction, and integration DLL deployment for versioned software updates.

Portfolio Website Live Site *JavaScript, React, CSS, Vite*

- Built a responsive portfolio website using React, JavaScript, and CSS. Implemented dynamic project sections with modal interactions and smooth animations, following a reusable component-based architecture with efficient state management. Optimized performance using Vite and ensured cross-device compatibility.